

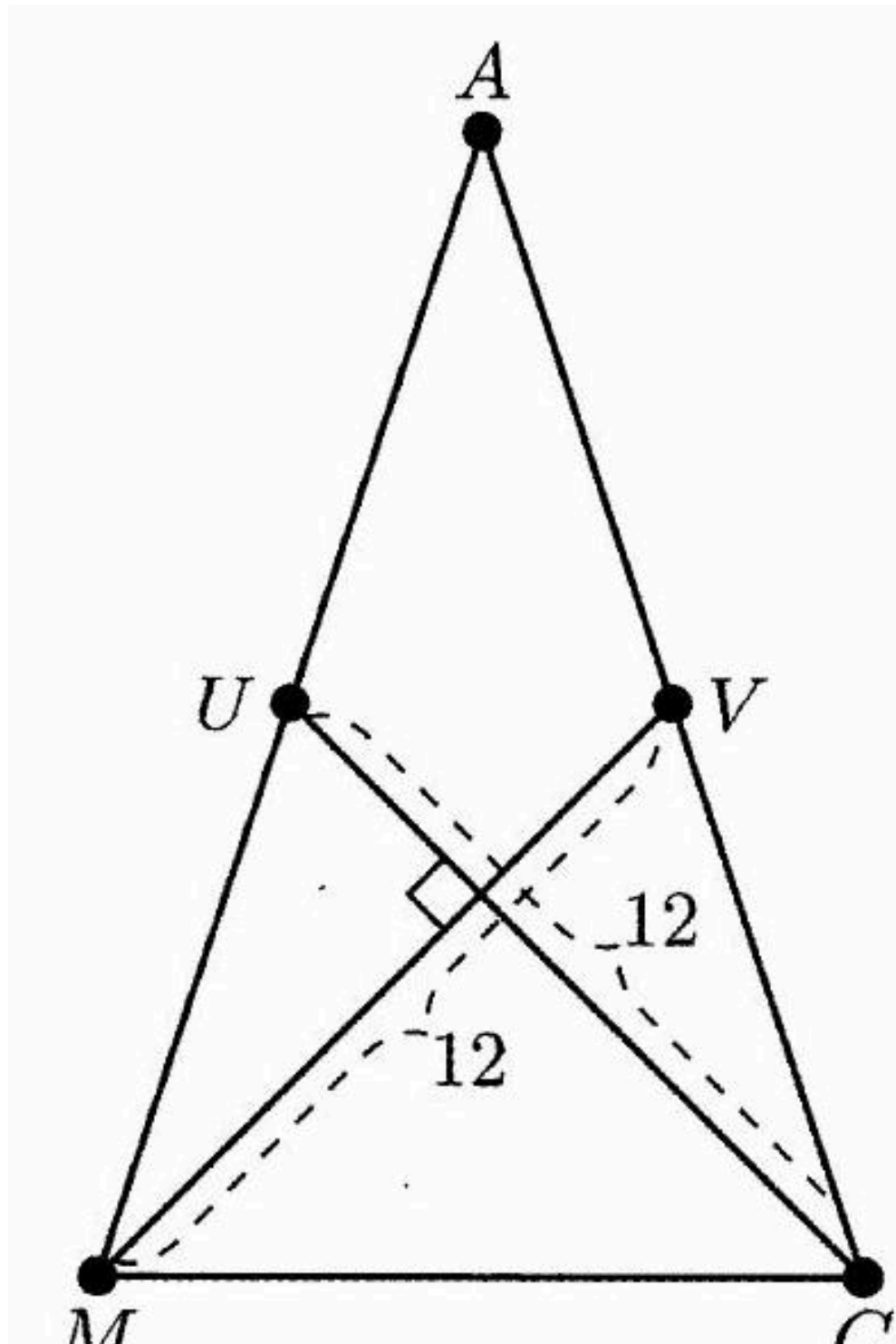
2020 AMC 10A Problem 12 - Solution

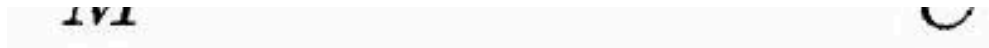
Review the full statement and step-by-step solution for 2020 AMC 10A Problem 12. Great practice for AMC 10, AMC 12, AIME, and other math contests

2020 AMC 10A Problem 12

Problem:

Triangle AMC is isosceles with $AM = AC$. Medians $\overline{MV}$ and $\overline{CU}$ are perpendicular to each other, and $MV = CU = 12$. What is the area of $\triangle AMC$?



**Answer Choices:**

- A. 48
- B. 72
- C. 96
- D. 144
- E. 192

Solution:

Since quadrilateral $UVC M$ has perpendicular diagonals, its area can be found as half of the product of the length of the diagonals. Also note that $\triangle AUV$ has $\frac{1}{4}$ the area of triangle AMC by similarity, so

$$[UVC M] = \frac{3}{4} \cdot [AMC]$$

Thus,

$$\begin{aligned} \frac{1}{2} \cdot 12 \cdot 12 &= \frac{3}{4} \cdot [AMC] \\ 72 &= \frac{3}{4} \cdot [AMC] \\ [AMC] &= \boxed{(C) 96}. \end{aligned}$$

OR

We know that $\triangle AUV \sim \triangle AMC$, and since the ratios of its sides are $\frac{1}{2}$, the ratio of of their areas is $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$. If $\triangle AUV$ is $\frac{1}{4}$ the area of $\triangle AMC$, then trapezoid $MUVC$ is $\frac{3}{4}$ the area of $\triangle AMC$.

Let's call the intersection of $\overline{UC}$ and $\overline{MV}P$.

$$\text{Let } \overline{UP} = x$$

$$\text{Then } \overline{PC} = 12 - x$$

Since $\overline{UC} \perp \overline{MV}$, $\overline{UP}$ and $\overline{CP}$ are heights of triangles $\triangle MUV$ and $\triangle MCV$, respectively. Both of these triangles have base 12.

$$\text{Area of } \triangle MUV = \frac{x \cdot 12}{2} = 6x$$

$$\text{Area of } \triangle MCV = \frac{(12 - x) \cdot 12}{2} = 72 - 6x$$

Adding these two gives us the area of trapezoid $MUVC$, which is

$$6x + (72 - 6x) = 72$$

This is $\frac{3}{4}$ of the triangle, so the area of the triangle is

$$\frac{4}{3} \cdot 72 = \boxed{(C) 96}.$$

The problems and solutions on this page are the property of the MAA's [American Mathematics Competitions](https://www.maa.org/math-competitions) 

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